

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 4 Summer 2025 (Regular)

Course : Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTCVC402 - ENVIRONMENTAL ENGINEERING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 The unit of turbidity of water is _____
- a) JTU
 - b) NTU
 - c) PPM
 - d) All of above

Dosage of alum is decided by _____ test.

- a) Hardness
- b) Chloride
- c) Jar
- d) Char

- 3 Cleaning interval for rapid sand filter bed is _____.
- a) 3 – 4 months
 - b) 6 – 8 months
 - c) 1 – 2 year
 - d) 24 hours

If the sedimentation tank is rectangular in shape having length L, width W and depth D, then for discharge (Q), the settling velocity of particle would be _____.

- a) $Q / (W \times D)$
- b) $Q / (L \times W)$
- c) $Q / (D \times L)$
- d) $Q / (L \times W \times D)$

- 5 BOD₅ is determined in laboratory at _____ degree celcius.

- a) 15 b) 20 c) 25 d) 10

- 6 Screen chamber is provided to remove _____ from raw wastewater

- a) Dissolved Solids
- b) Volatile Solids
- c) fixed solids
- d) Suspended Solids

- 7 Ozone is present in _____ layer of Atmosphere
 a) troposphere
 b) mesosphere
 c) stratosphere
 d) thermosphere
- 8 Pick out the odd one with respect to classification of solid wastepaper. card board, wood, metal
 a) paper
 b) card board
 c) wood
 d) metal
- Function of cascade aerator is to _____.
 a) remove iron and manganese
 b) to remove odors
 c) to increase oxygen content in water
 d) to mix chemicals if added in water e) all of above
- 10 _____ condition is most unfavourable for dispersion of air pollutants in the atmosphere. a) Inversion b) unstable c) Neutral d) all of above
- 11 The ultimate (Last) disposal option for solid waste management is _____. a) segregation b) Composting c) Land filling d) none of above

- As per IS 10500 MPN should be _____.
 a) zero
 b) 10
 c) 20
 d) 100

Q2. Solve the following.

- A) List population forecasting methods. Explain arithmetic mean method. The population of a town was 25,000 in 1991, 36,000 in 2001, and 47,000 in 2011. Estimate the population in 2031 using the **Arithmetic Increase Method**. 6
- B) Explain various types of water demands? Write fire demand formulae 6

Q3. Solve the following.

- A) Design a clariflocculator to treat a flow of 10 MLD. Assume necessary data 6
- B) What is meant by back-washing? Explain how cleaning is performed in Rapid sand Filter with neat sketch 6

Q4. Solve Any Two of the following.

- A) List out four common types of water distribution system layouts. Discuss any one in detail with neat diagram. 6
- B) Compare **gravity, pumping, and combined (gravity + pumping)** methods of water supply. Which is most preferred and why? 6

- C) Three pipes are connected **in series** to carry water between two reservoirs. Their **lengths and diameters** are as follows: 6

Pipe	Length (L) in meters	Diameter (D) in mm
A	400	300
B	600	250
C	500	200

Assume that the **friction factor (f)** is the same for all pipes, and that the **Darcy-Weisbach equation** applies.

Calculate the **equivalent length of a single pipe** of diameter **200 mm** that will produce the same head loss as the three pipes in series.

- Q5. Solve Any Two of the following.

- A) Draw a wastewater treatment flowsheet and tabulate functions of each unit. 6
- B) Define 'Solid Waste'. Classify the following solid wastes as **biodegradable or non-biodegradable**. Also suggest disposal methods for each category. 6
Items: 1) Food waste, 2) Plastic bottles, 3) Vegetable peels, 4) Broken glass, 5) Paper
- C) Define **BOD**. A wastewater sample was tested for BOD. The 5-day BOD at 20°C was found to be **120 mg/L**. If the reaction rate constant K_D at 20°C is **0.1 day⁻¹**, calculate the **ultimate BOD (BOD_u)** using the first-order BOD equation. 6

- Q6. Solve Any Two of the following.

- A) Define **air pollution** and classify the **types of air pollutants** with suitable examples. 6
- B) Explain **atmospheric stability** and its effect on **plume behavior**. Sketch and label any **five common plume types**. 6
- C) List **air pollution control devices** used for industrial air emissions and state working principle, advantages and disadvantages of **gravity settling chamber** with neat sketch. 6

*** End ***